RAJALAKSHMI ENGINEERING COLLEGE

Thiru Selvam R K 250701791-2025-CSE

T2

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|---|---|
| **Started on** | Sunday, 30 August 2026, 11:08 PM |
| **State** | Finished |
| **Completed on** | Sunday, 30 August 2026, 11:48 PM |
| **Time taken** | 40 mins 18 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

**Example 1:**

```
Input: nums = [3,2,3]
Output: 3
```

**Example 2:**

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

**Constraints:**

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

**For example:**

| Input | Result |
|---|---|
| 3<br>3 2 3 | 3 |
| 7<br>2 2 1 1 1 2 2 | 2 |

**Answer:** (penalty regime: 0 %)

```
#include <stdio.h>

int majority(int a[], int l, int r)
{
    if (l == r)
        return a[l];

    int mid = (l + r) / 2;

    int left = majority(a, l, mid);
    int right = majority(a, mid + 1, r);

    if (left == right)
        return left;

    int c1 = 0, c2 = 0;

    for (int i = l; i <= r; i++)
    {
        if (a[i] == left)
            c1++;
        if (a[i] == right)
            c2++;
    }

    if (c1 > (r - l + 1) / 2)
        return left;

    return right;
}

int main()
{
    int n;
```

```
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("%d", majority(a, 0, n - 1));

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>3 2 3 | 3 | 3 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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